

Introduction to Computational Thinking and Python Programming

Instructor: Sarom Leang, PhD (Teacher/Professor)

Specialist: Crystal Rafael-Castelan

Innovation Hall 203



Pronouns

- Instructor: Sarom Leang, Ph.D. (Teacher/Professor)
- Specialist: Crystal Rafael-Castelan



Schedule

- 10:00 AM – 10:15 AM (Homeroom)
- 10:15 AM – 11:35 AM (G1)
- 11:40 AM – 12:35 PM (Lunch)
- 12:40 PM – 02:00 PM (G2)



Student Expectations

- **NO FOOD**
- **NO DRINKS** (on the table)
- Be respectful to individuals and property
- Be open to learning
- Be open to not understanding
- Be patient with yourself
- Ask questions
- Explore
- **Embrace failure**



Course Overview (cont.)

Exploratory Topics

- ~~Session #1 – November 1, 2025~~
 - ~~Entrance Survey~~
 - ~~How Computers Work~~
- Session #2 – December 6, 2025
 - Parallel Computing
- Session #3 – January 31, 2025
 - Generative Artificial Intelligence
- Session #4 – March 7, 2025
 - Quantum Computing
- Session #5 – April 11, 2025
 - Future of Computing
 - Exit survey

Topics

- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals



Entrance Survey (Results)

Question	G1	G2
Have you taken this course before?	1 Yes, 14 No	2 Yes, 14 No
Current grade level?	0, 4, 7, 3, 1	2, 6, 5, 3, 0
Do you have access to a laptop or computer at home?	14 Yes, 1 No	16 Yes, 0 No
Do you have access to a laptop or computer at school?	100% Yes	100% Yes
Do you have any experience in computer programming?	7 Yes, 8 No	8 Yes, 8 No
Keyboard is an input device	1 wrong	1 wrong
CPU is known as the brain of a computer	2 wrong	1 wrong
Speaker is an example of an output device	0 wrong	3 wrong
All the above are storage devices	0 wrong	3 wrong
Storage device can hold data and programs even if the power is off	0 wrong	2 wrong
Internet is a worldwide collection of networks	2 wrong	4 wrong
A supercomputer is the fastest, most powerful computer	3 wrong	4 wrong
Data is raw, unprocessed facts including text, numbers, images, and sound	2 wrong	4 wrong
Computer can execute code in machine language	10 wrong	11 wrong



Use of Class Time

- Course is capped at 50 seats
- Each seat costs \$50
- EIP has invested \$2500 to offer this course
- Bring in schoolwork to occupy yourself



Ozaria Progress Winners – \$50 Visa Gift Card

- **G1:** Marion M
- **G2:** Aaron C



Supercomputing Conference



HPC = high performance computing



High Performance

Example	General	High	Metrics
Cars	Consumer e.g., Ford Fusion	Performance e.g., Formula-1, Koenigsegg Agera	• Speed • Torque (force) • Horse-power (force x distance)
Computers	Consumer e.g., Alienware Area 51	Performance e.g., HPE Cray El Capitan	• Math operations per second • Number of CPUs • Aggregate memory



High Performance Computing

Specs	Alienware Area 51	HPE El Capitan (11,136 nodes) #1
CPU	1x AMD Ryzen 2950X 16-core/32-threads 4.4 GHz	4x AMD MI300A APUs (CPU+GPU) 24-core Zen4 x86 CPU per APU
GPU	2x NVIDIA RTX 2080 Ti	1x AMD CDNA3 GPU per APU
Memory	64 GB	512 GB
Floating point operations per second (FLOPS)	700 GFLOPS	250,800 GFLOPS (per node) 250.8 TFLOPS (per node) 358x 2,792,908,800 GFLOPS (total) 2,792,908 TFLOPS (total) 3,990x



FLOPS

- FLOPS stands for **F**loating-point **O**perations per **S**econd
- Floating-point means a number with a decimal
- An operation means a mathematical operation (add or multiply)
- 1 FLOP = 1 floating-point operation
- 1 FLOPS = 1 floating-point operation per second
- GFLOPS = giga-FLOPS or $\times 10^9$ FLOPS
- TFLOPS = tera-FLOPS or $\times 10^{12}$ FLOPS



Top 5 Supercomputers (November 2025)

Rank	Country / System (CPU, GPU, Network, Vendor)	Rpeak (PFlops/s)
1	USA / El Capitan AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, HPE	2,821
2	USA / Frontier AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE	2,055
3	USA / Aurora Intel Xeon CPU Max 9470 52C 2.4 GHz, Intel Data Center GPU, Slingshot-11, Intel	1,980
4	Germany / JUPITER Booster NVIDIA GH200 Superchip 72C 3GHz, NVIDIA Infiniband NDR, EVIDEN	1,226
5	USA / Eagle Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft	846

Giga, Tera, Peta, Exa

Unit Prefix	Factor
Giga	1,000,000,000
Tera	1,000,000,000,000
Peta	1,000,000,000,000,000
Exa	1,000,000,000,000,000,000

2,800 PetaFLOPS = 2.8 ExaFLOPS

2,800,000,000,000,000,000
FLOPS

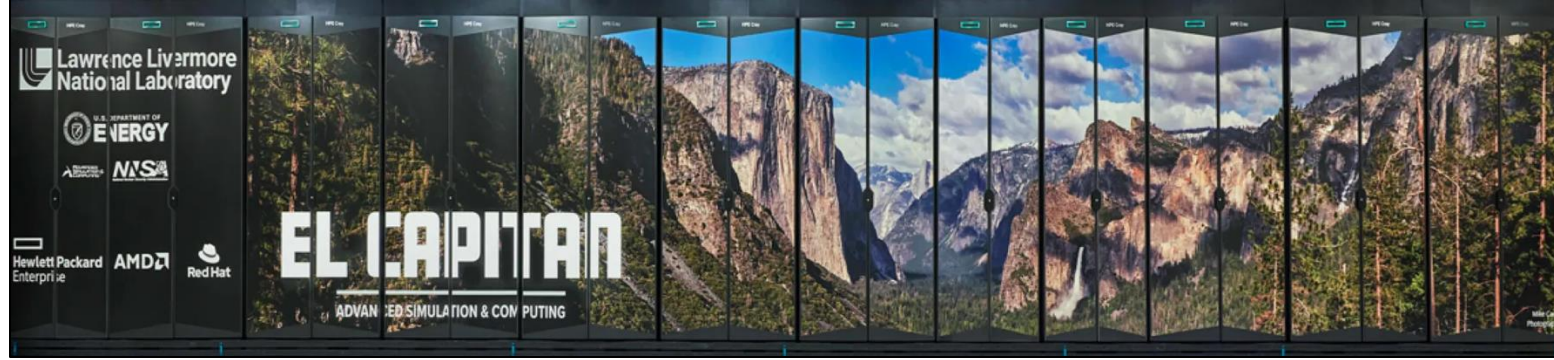


What Does a Supercomputer Look Like?



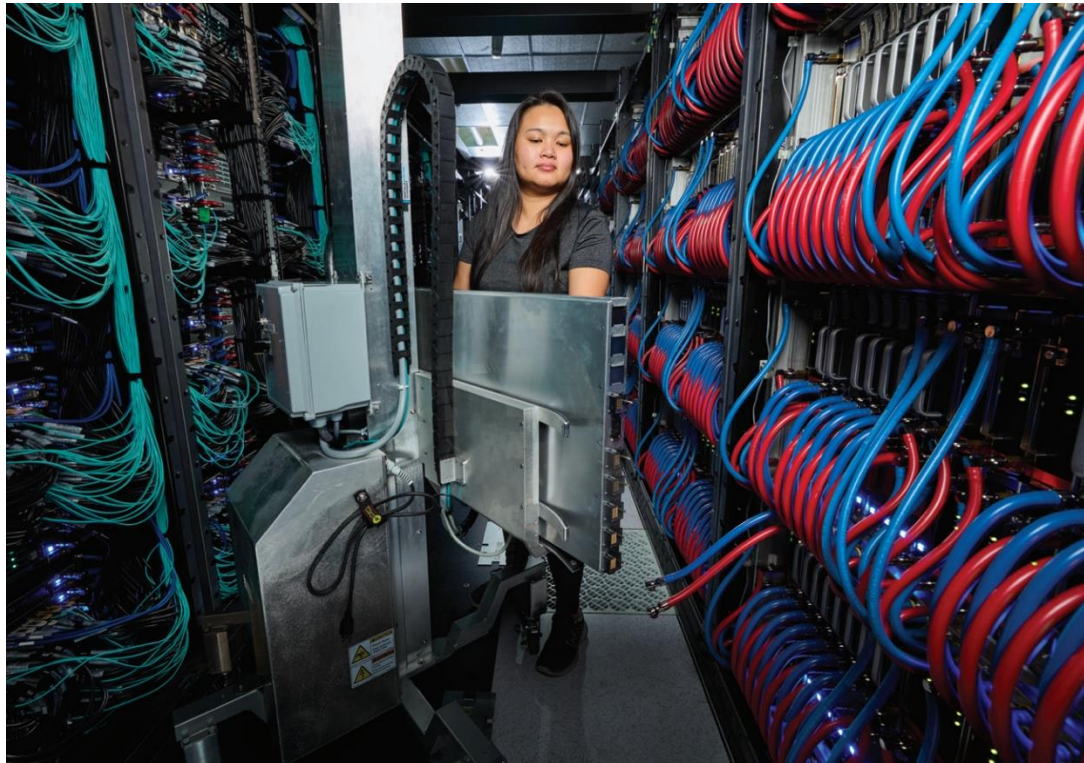
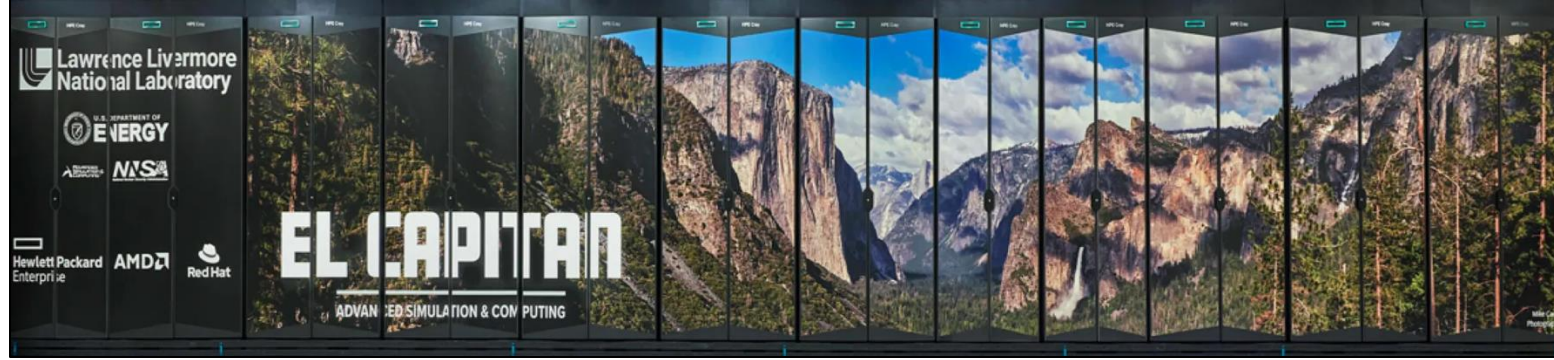
Top 5 Supercomputers





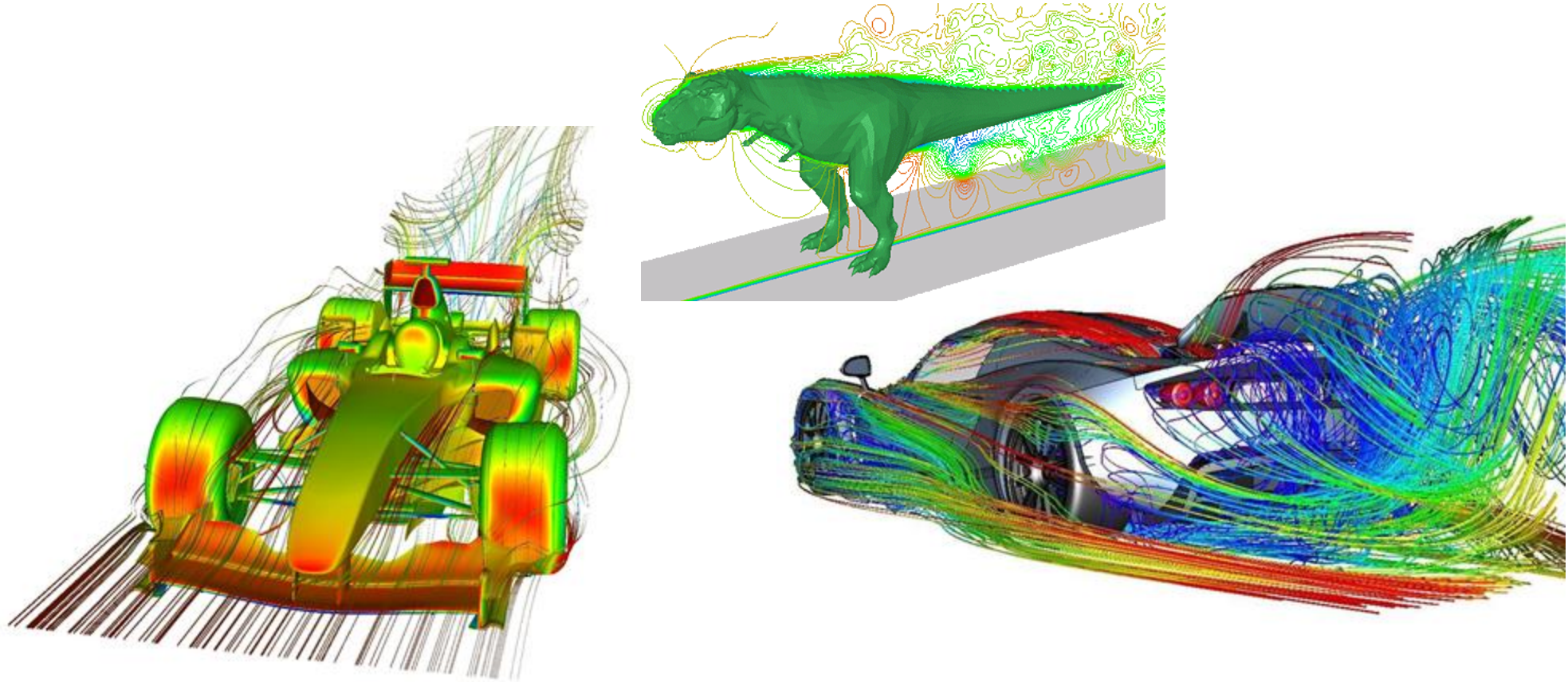
- If you went back in time **2.8 quintillion seconds**, you'd arrive over **70 billion years before** the Big Bang
- Equal to **one million** of today's **fastest smartphones** working on a calculation simultaneously
- Requires **5 to 9 million gallons of water** (8 to 14 Olympic-size pools) for cooling circulation (**every 24 hours**)





- If every one of the world's **8 billion people** worked on a calculation, **every second** of **every day**, around the clock, it would take **8 years** to do what El Capitan can do in **one second**
- Required the installation of **32 miles** of power/electrical cables

Supercomputing Applications



Computational Fluid Dynamics (CFD)

Supercomputing Applications

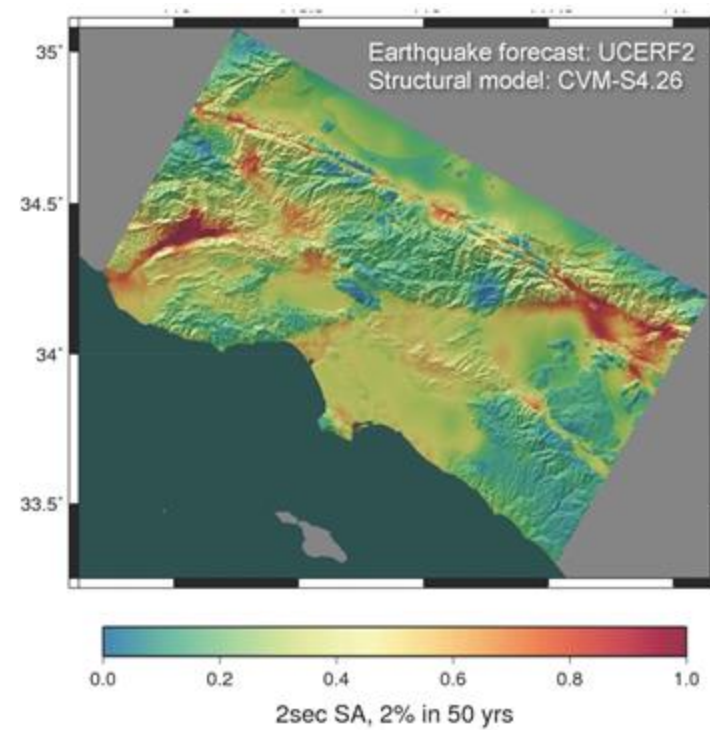
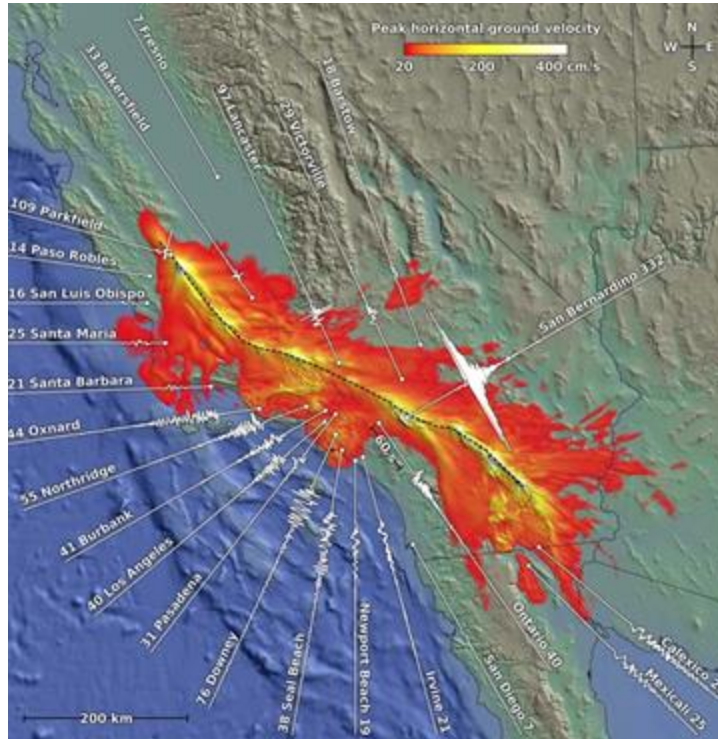


Computational Fluid Dynamics (CFD)

VIDEO

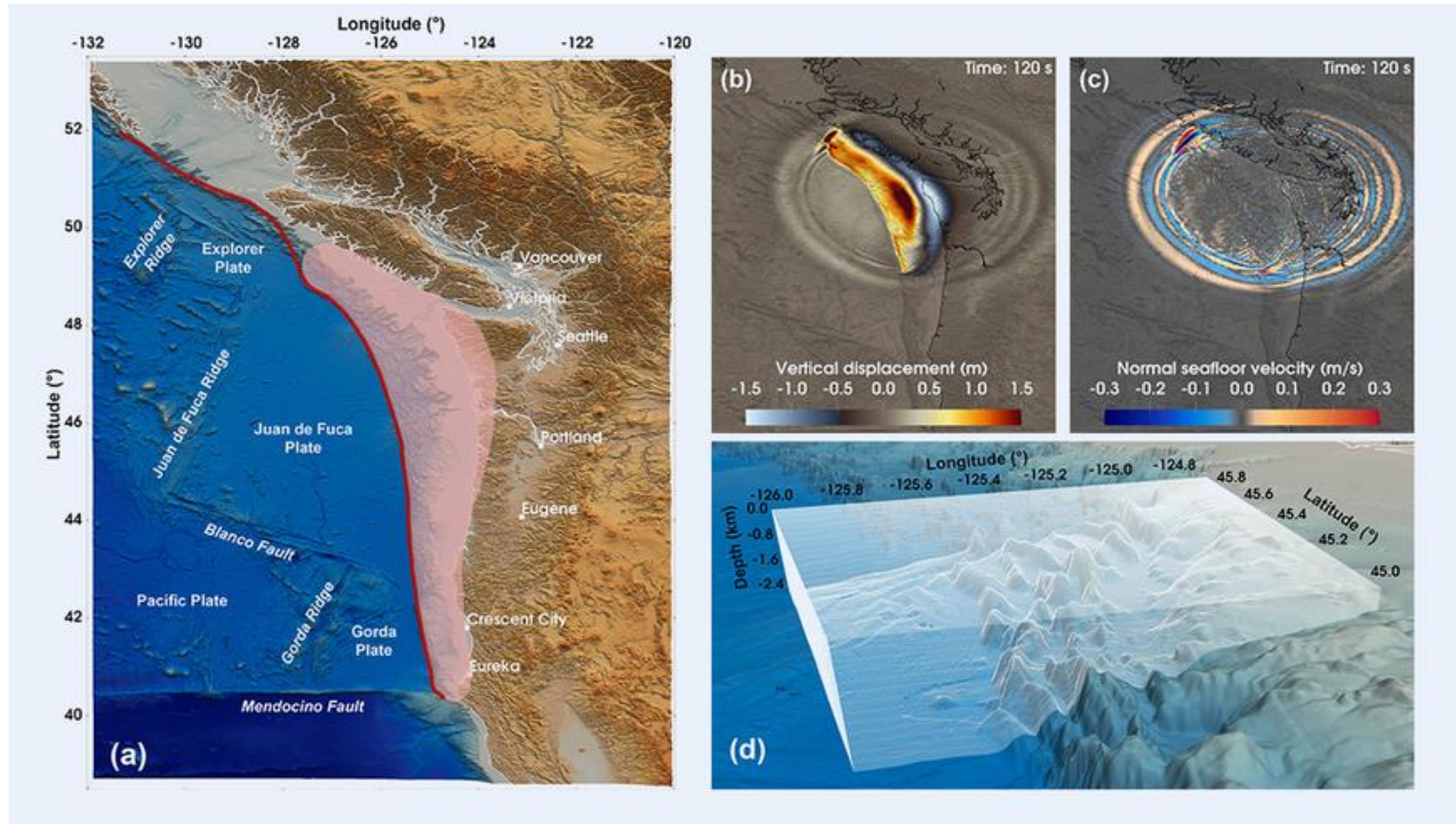


Supercomputing Applications



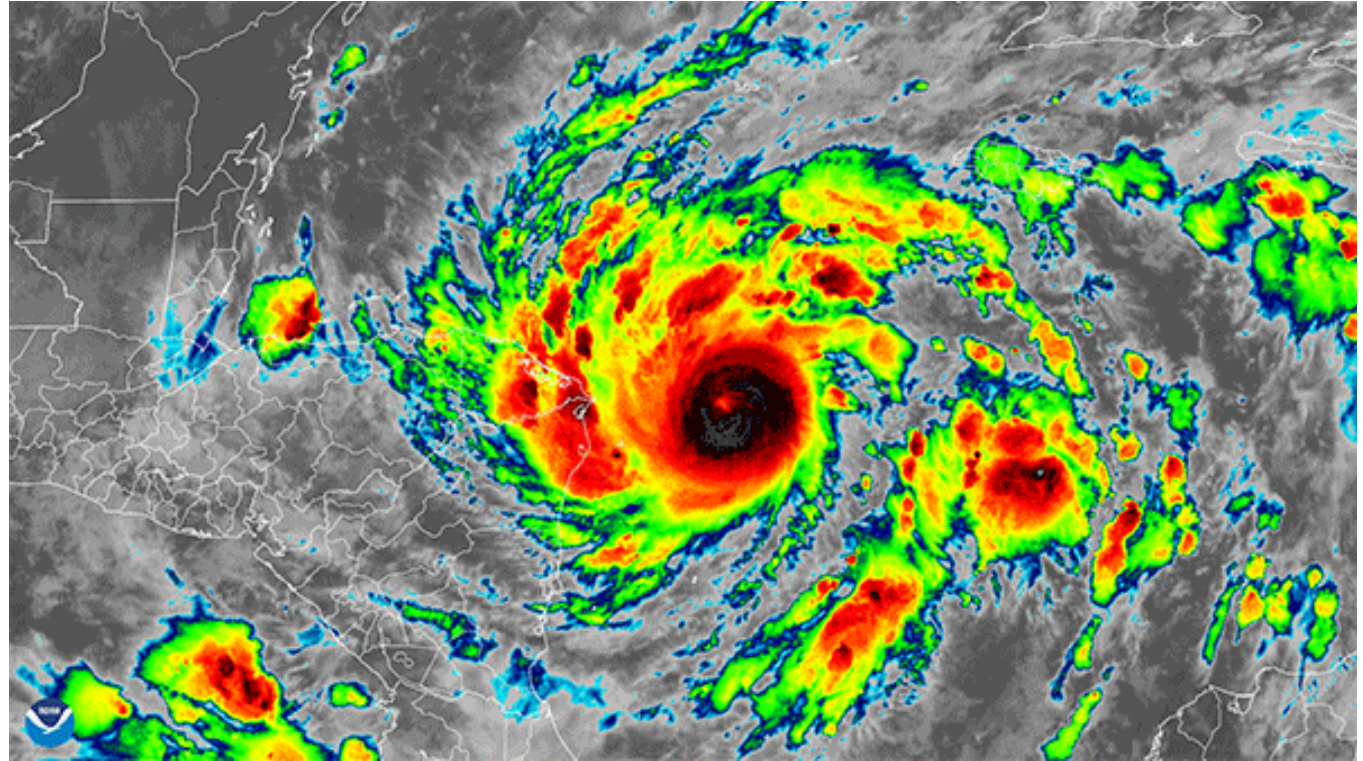
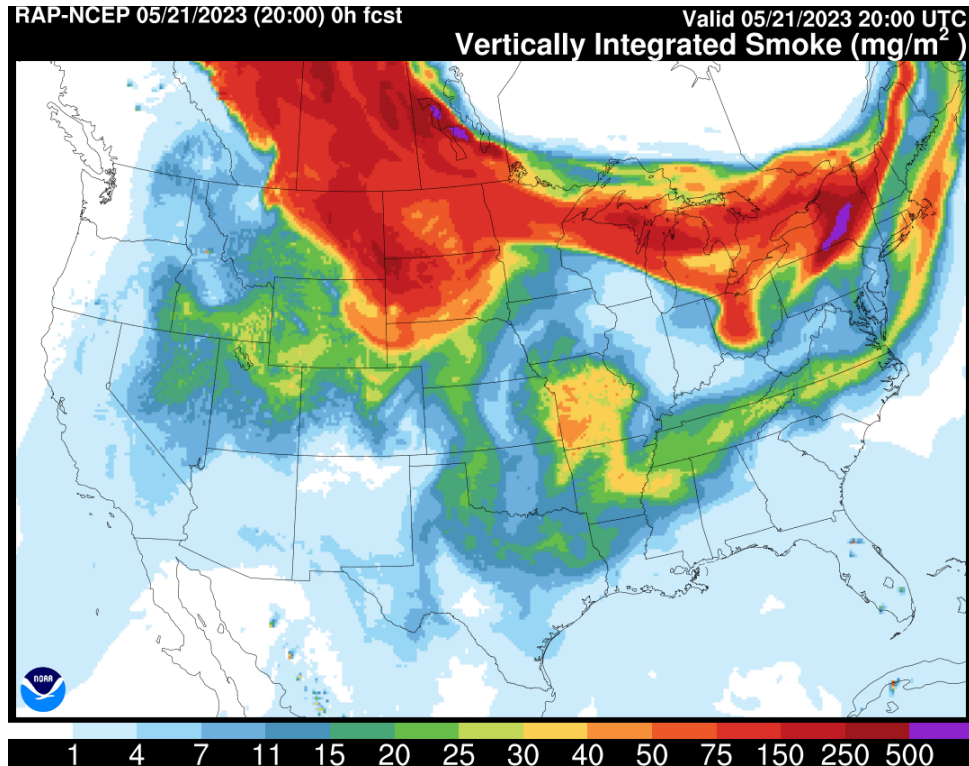
Seismic Modeling

Supercomputing Applications



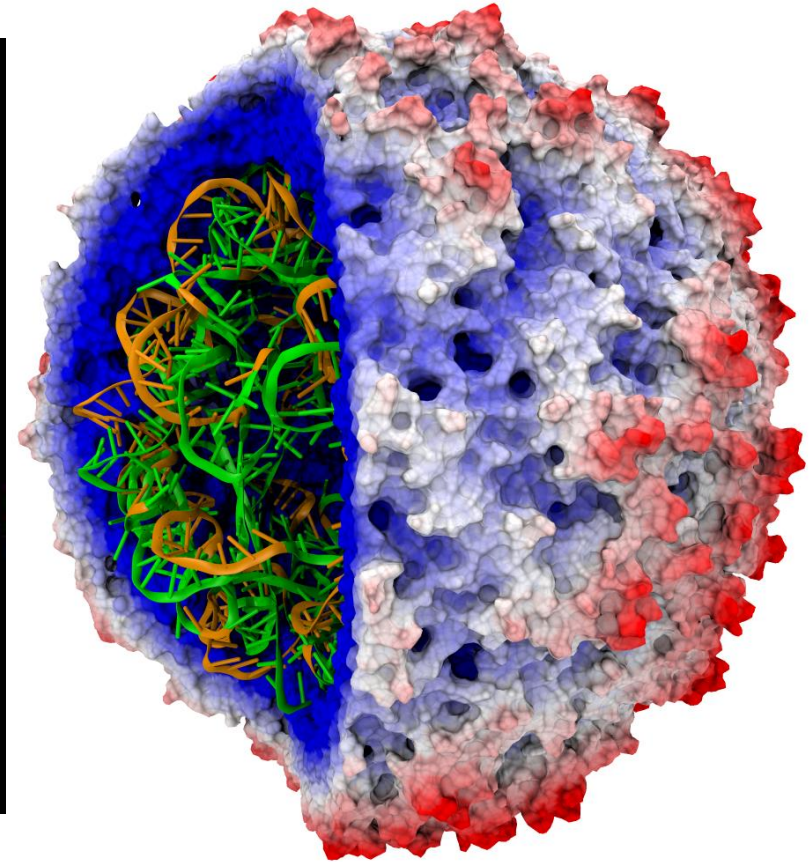
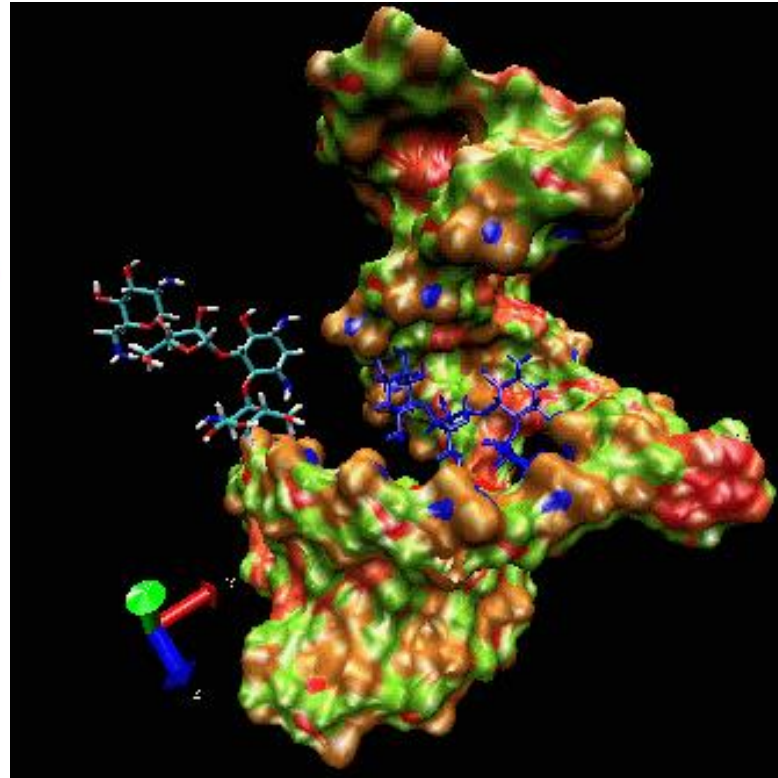
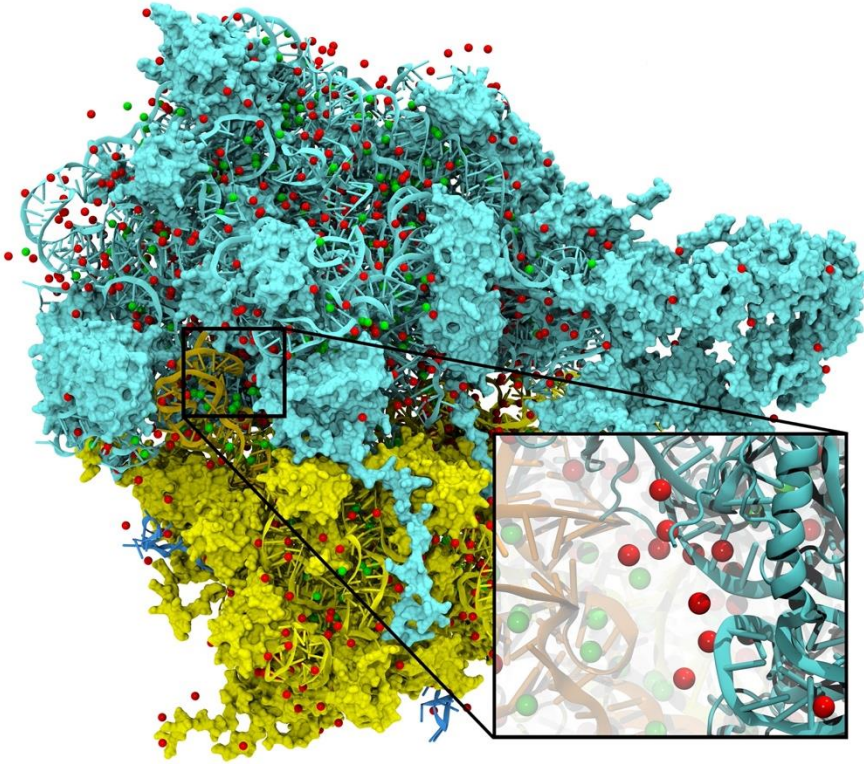
Tsunami Simulations

Supercomputing Applications



Weather Modeling

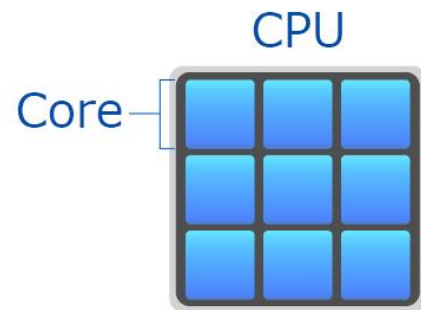
Supercomputing Applications



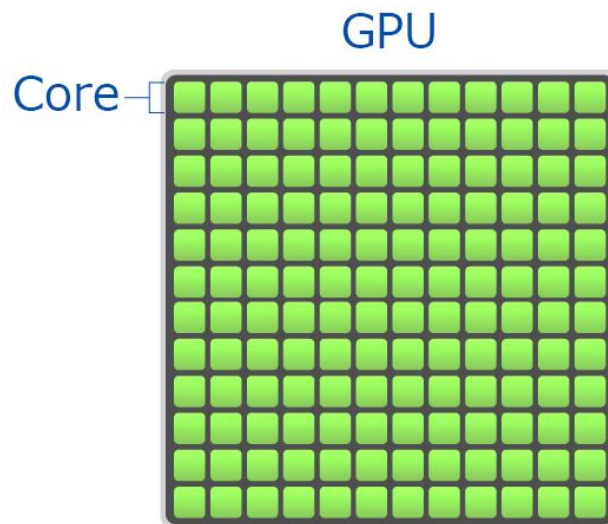
Molecular Simulations

Artificial Intelligence (AI) / Machine Learning (ML)

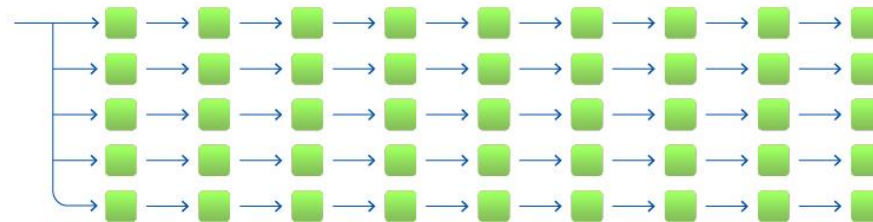
Sample applications but on General Purpose GPUs



The CPU performs continuous processing

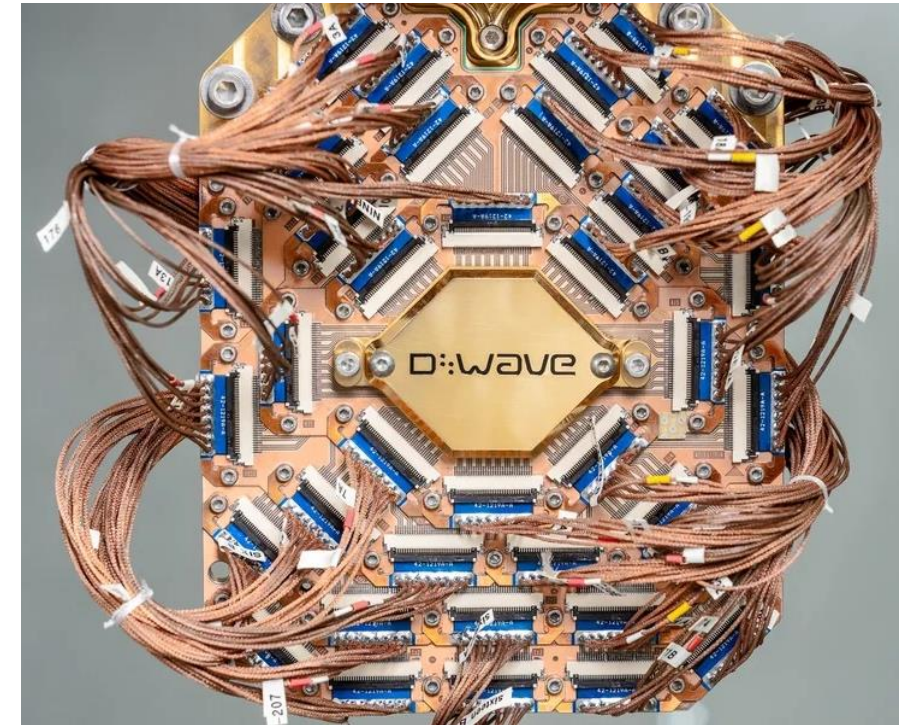
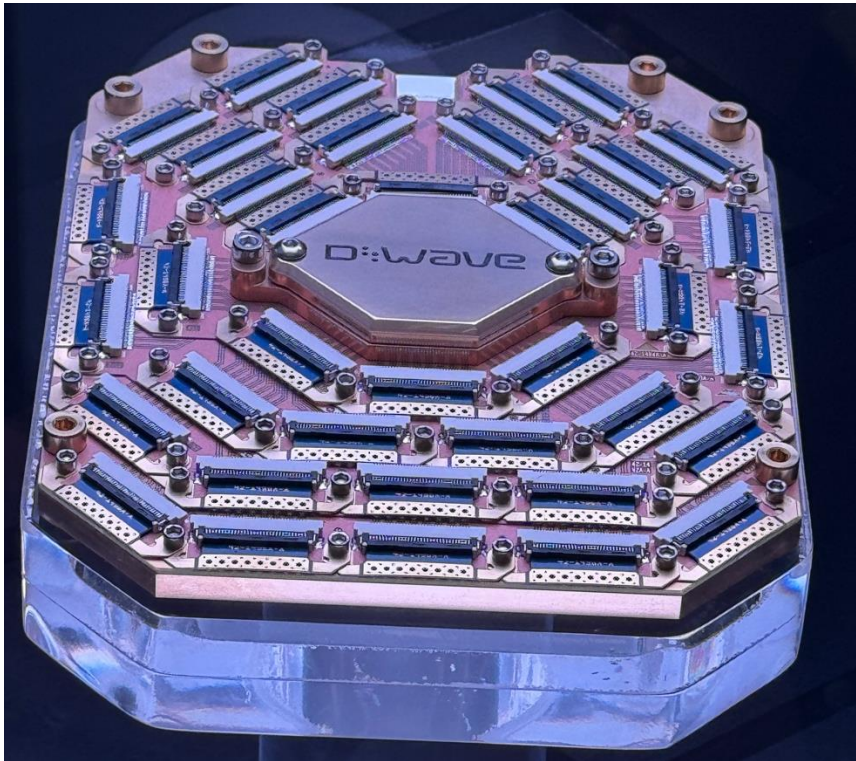


The GPU performs parallel processing



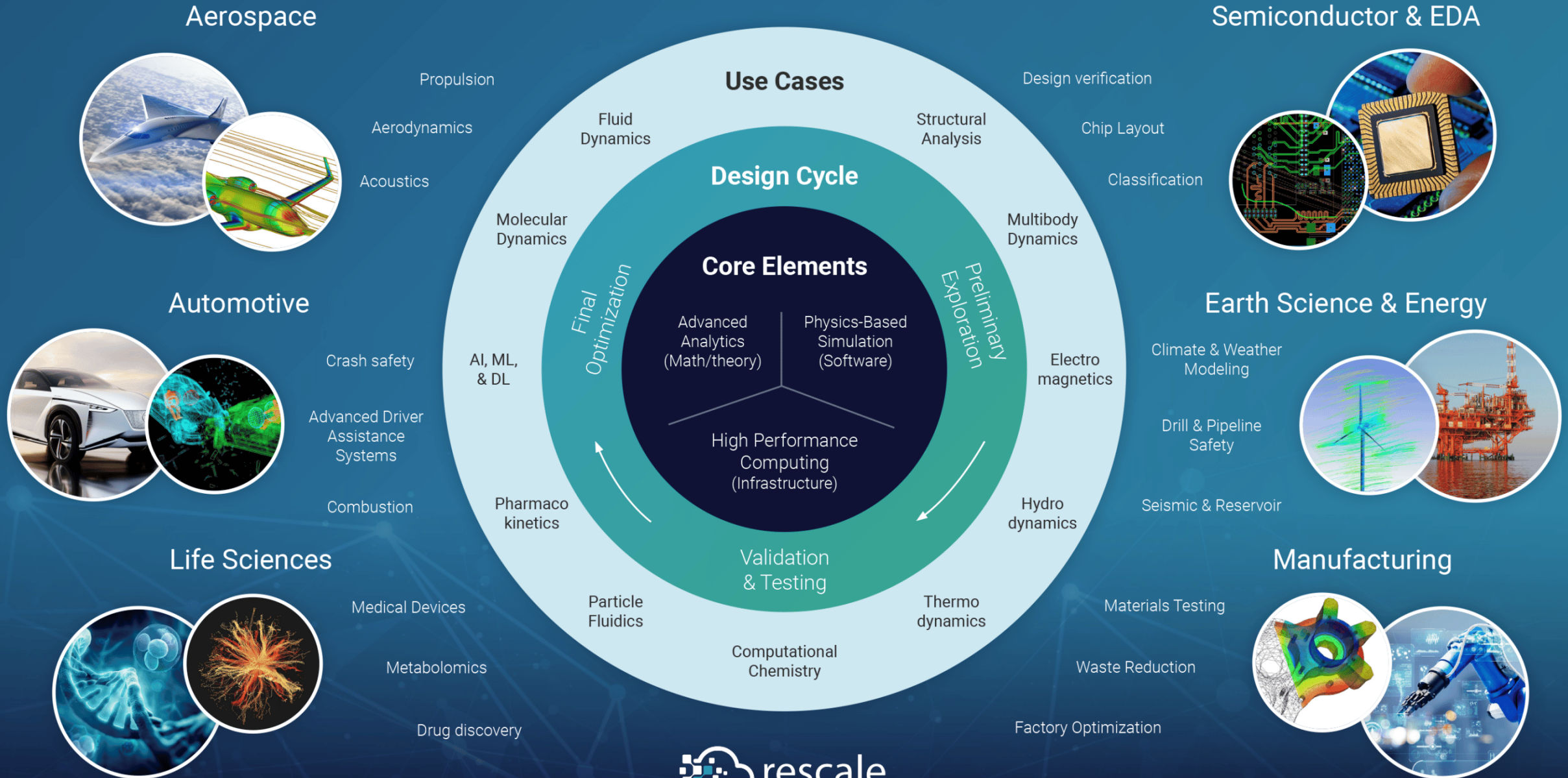
Quantum Computing (Hybrid/Simulations)

A new paradigm of computing



COMPUTATIONAL SCIENCE & ENGINEERING

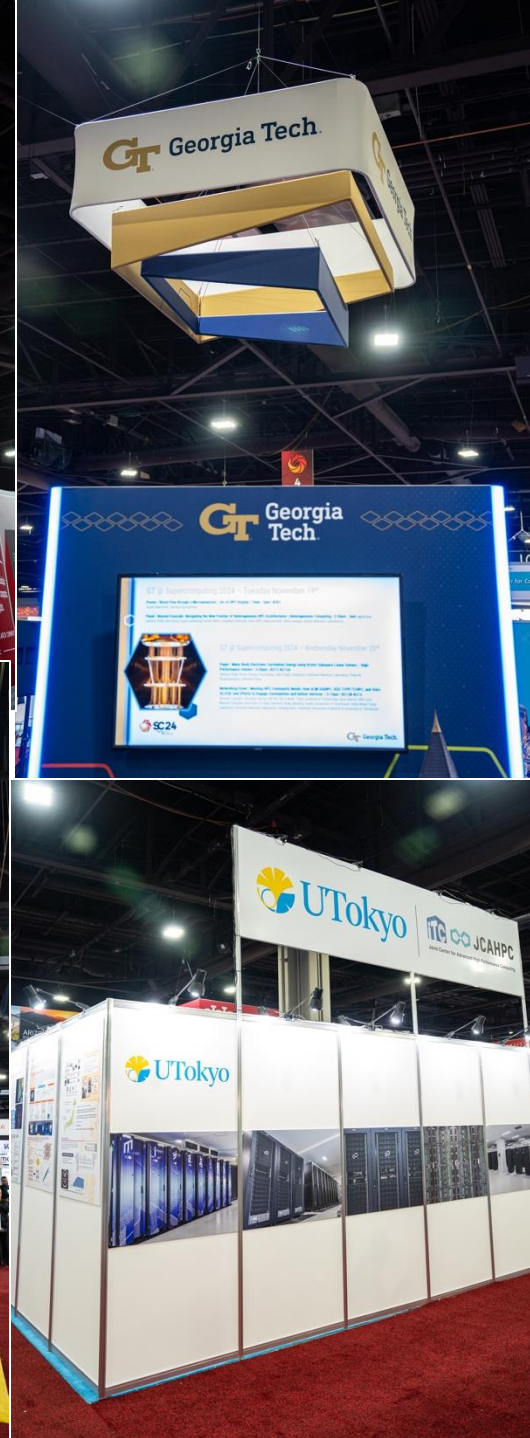
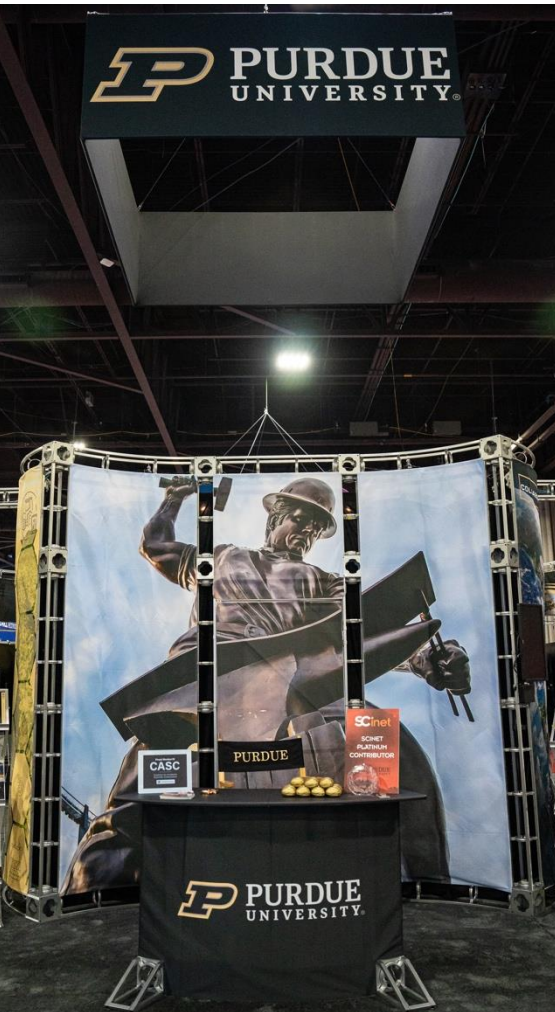
Building Blocks



Who is Interested in HPC?



Who is Interested in HPC?



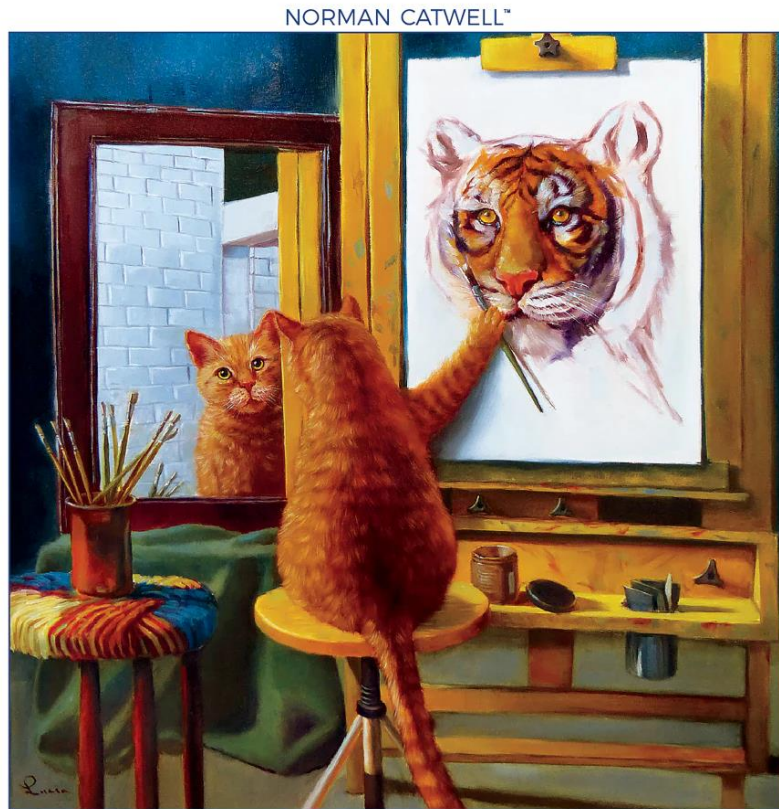


SC26

Chicago,
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Computational Thinking – 2D Puzzle



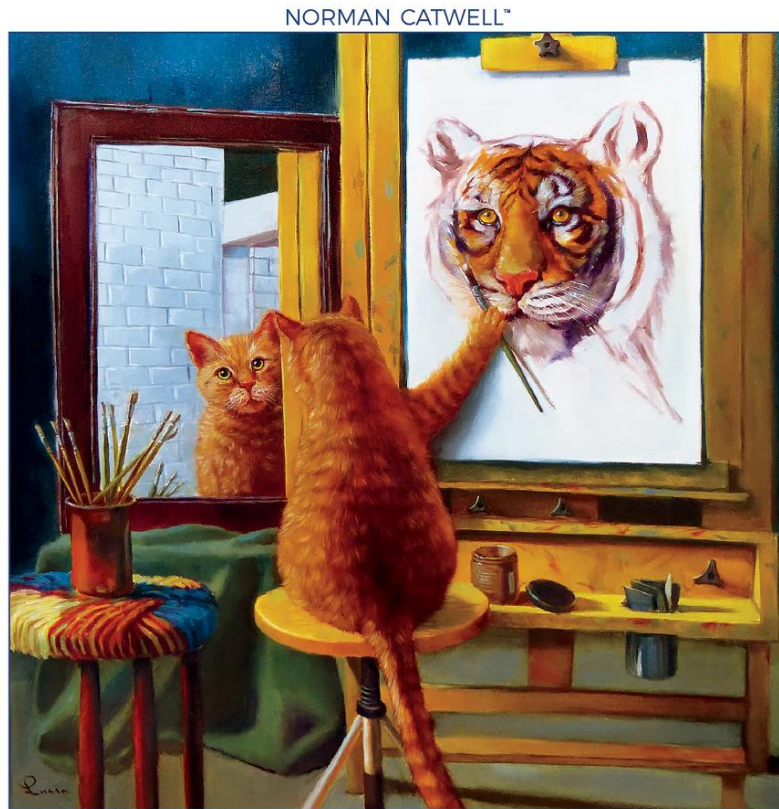
 300
PUZZLE | ROMPECABEZAS



- **Problem** – connect the greatest number of puzzle pieces together
- How can we break down this problem into smaller tasks?
- Any similarities between these smaller tasks?
- Can we generalize the similar tasks?
- Can we develop a step-by-step solution/strategy to the problem?



Computational Thinking – 2D Puzzle



300
PUZZLE | ROMPECABEZAS



- **Problem** – connect the greatest number of puzzle pieces together
- **Decomposition** – preparation, classify pieces, build frame, sub-region assembly, final integration
- **Pattern Recognition** – edge vs interior, color/texture, shape
- **Abstraction** – piece type (edge or interior), shape, color/texture, target region
- **Algorithm Design** – see next slide



Algorithm

1. Initialize workspace
2. Expose all pieces
3. Sort by visual features
4. Assemble the frame
5. Solve distinct regions
6. Refine using shape (ambiguous pieces)
7. Repeat until complete
8. Verify





Exercise – Human Supercomputer?

- **Challenge:** 300 piece 2D jigsaw puzzle
- **Goal:** Get the most number of correctly connected pieces within the established time frame
- We will assign groups of (or close to):
 - 1
 - 2
 - 4
 - 8
 - 16



Before You Begin

- Spend 5 minutes to strategize your approach.
- Assign 1 person as the leader of your group.
- The group leader will assign tasks the other members of the group.



Exercise – Human Supercomputer?

- **Challenge:** 300 piece 2D jigsaw puzzle
- **Goal:** Get the most number of correctly connected pieces within the established time frame
- We will assign groups of (or close to):

- 1
- 2
- 4
- 8
- 16

# Solvers	G1 (# connected)	G2 (# connected)
1		
2		
4		
8		
16		



Discussion – Human Supercomputer

- Expected results?
- Approaches/strategies?
- Benefits of working in your environment?
- Challenges of working in your environment?
- What do you expect to happen to the results if the number of puzzle pieces increased (more puzzle pieces)?
- What do you expect to happen to the results if the number of puzzle pieces decreased (less puzzle pieces)?



Challenges in High Performance Computing

- **Large** enough **problem size**
- Break up the problem into **independent tasks**
- **Distributing/assigning tasks** to available compute units
- Creating algorithms that **minimizes** the amount of **communication**

